Cleaning Validation Protocol of ASYM-140851

(Batch Size: 157.69 ~ 292.80 kg (Output)

At Asymchem Life Science (Tianjin) Co., Ltd.

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Revision History

Protocol No.	Effective Date	Document Changes	Justification
CVP-1400210301.01		The first version	N/A
Note1: The document shall only take effect after the training is completed, and the effective date will be manually filled in by designated personnel with signature and date once confirmed			

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1. Purpose

The purpose of this cleaning validation protocol is to ensure that the cleaning program of all the material-contact equipment used for manufacturing ASYM-140851 (CPo140021-03, Step 1) are in place and that product contamination is minimized.

2. Scope

This protocol is applicable for cleaning validation of the equipment used for manufacturing step 1 in-process product ASYM-140851 according to the defined process (see PVP-1400210301 <Process Performance Qualification Protocol of ASYM-140851>(current version)) in the defined equipment train (**Fig.1**) and the defined batch size (157.69~ 292.80kg) in Building 12 Module 13 at Asymchem Life Science (Tianjin) Co., Ltd (Address: No. 265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China, abbr.: Asymchem TJ4). The cleaning validation will be performed following the approved cleaning method (CMa-TJ414002101<Equipment Cleaning Method > (current version), CMb-TJ414002101<Equipment Cleaning Method > (current version). The cleaning validation will cover at least three continuous successful cleaning verification cycles.

The production building/module/equipment used for the manufacturing of the process intermediate ASYM-140851 are non-dedicated.

To sum up, the equipment and cleaning information involved in this Cleaning Validation Protocol is listed in **Table 1**. Detail evaluation of each specific part are further determined in separate section.

Table 1, The equipment and cleaning information involved in this Cleaning Validation Protocol

Equipment name	**Equipment ID**	**Cleaning Process**	**Cleaning frequency**		
			Batch to batch flush	**In-situ minor cleaning**	**Disassemble major cleaning**
Solid Phase Peptide Synthesizer	TJ4S-1213-Sy16 TJ4S-1213-Sy16-R01	Refer to CMa-TJ414002101<Equipment Cleaning Method > (current version), CMb-TJ414002101<Equipment Cleaning Method > (current version) which are evaluated during cleaning process design and development	Not applicable	Not applicable	After every batch
Agitation tank	TJ4S-1213-AT09		After each batch	Not applicable	After end of production
Agitation tank	TJ4S-1200-AT01		After each batch	Not applicable	After end of production

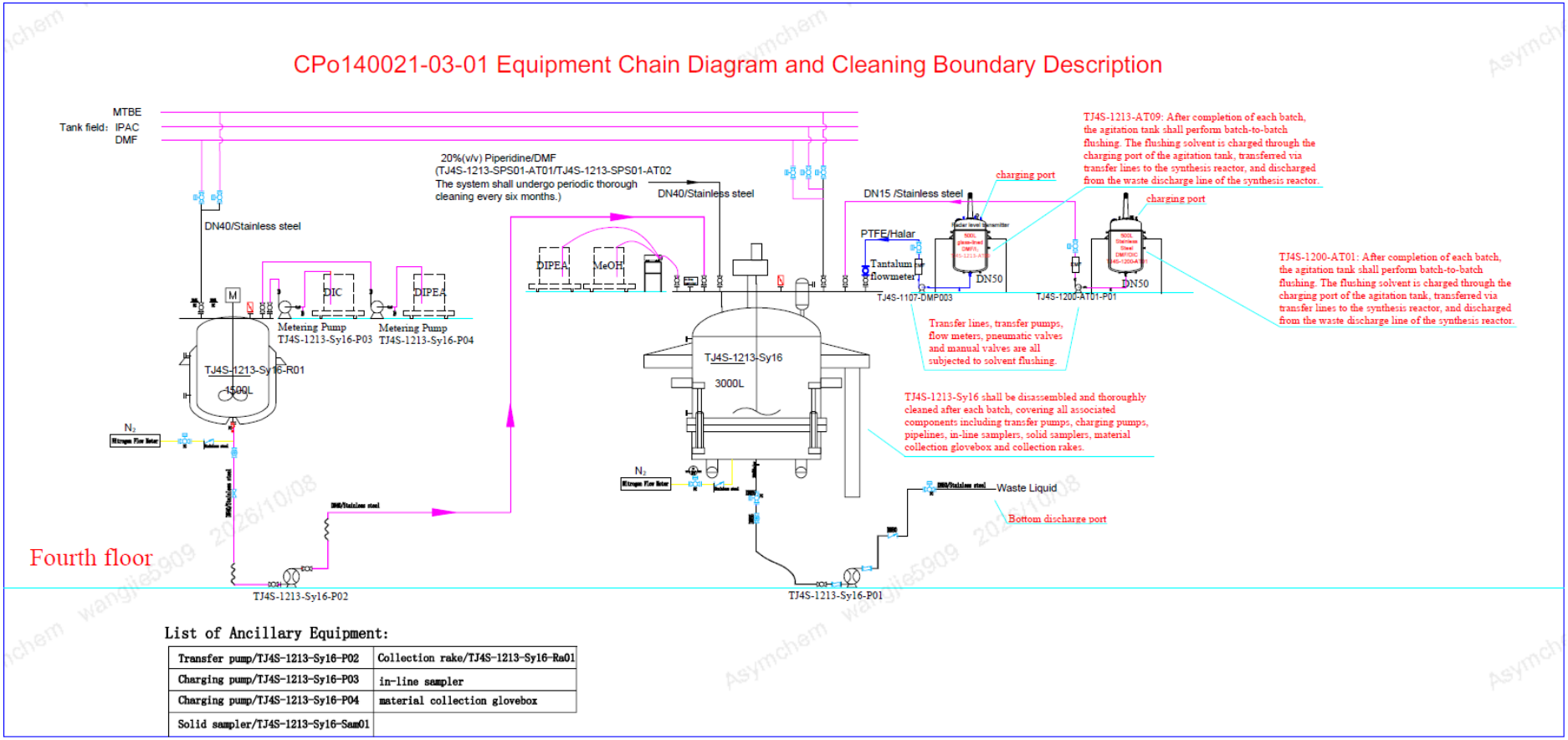


Fig. 1, The defined equipment train diagram and cleaning boundary

3. Responsibility

The following responsibilities are assigned to the relevant departments and as listed in **Table 2**.

Table 2, Functions and Responsibilities

Function	Responsible
Project Manager or Director	Prepare the cleaning validation protocol and report
QA engineer/ manager/director	Release the equipment after satisfactory cleaning Review the cleaning validation protocol and report
Production Manager/ director	Organize the Cleaning and collect the rinse/reflux sample
Director/ senior director/Associate Chief Engineer, Production	Review the cleaning validation protocol and report
Analytical department	Perform the swab sampling Analysis and reporting the analytical results to QA
Senior Director, QC	Approve the cleaning validation protocol and report
Executive Director, Chemical Macromolecule Division	Approve the cleaning validation protocol and report
VGM, Analysis of CMMD	Approve the cleaning validation protocol and report
The head of Production management department	Approve the cleaning validation protocol and report
The head of Quality management department	Approve the cleaning validation protocol and report

All the employees who participate in this cleaning validation will be trained on this protocol, the corresponding cleaning method and SOPs outlined in section 4 of this protocol. Responsible personnel shall enter his/her name and sign in <Sign in table> (QF06006), and attach this table to the cleaning validation report.

4. Procedure support

The corresponding SOPs are listed in **Table 3**.

Table 3, Corresponding SOP related to the cleaning validation, Equipment Cleaning, sampling and testing

Title	SOP version #
Cleaning Validation Master Plan	CVMP-01 (current version)
Equipment cleaning management procedure for GMP production	DS-GMP-003 (current version)
Cleaning Validation	DS-GMP-043 (current version)
OOS Result Investigation	QA-015 (current version)
ATR result investigation	QA-032 (current version)
Sampling and Testing Procedure for cleaning verification	AC-053 (current version)
Production Equipment Cleaning manual	QM03024 (current version)
Minimum lot size, rinse volume, reflux volume and internal surface area of equipment	QM05004 (current version)
Equipment to be swab-tested and the sampling point diagram	QM04005 (current version)

5. Cleaning Validation Strategy

According to Asymchem's SOP <Cleaning Validation> DS-GMP-043 (current version), the life cycle of cleaning validation includes three stages:

1) Stage 1: Cleaning process design and development, according to the pharmacology, toxicology, solubility and other information obtained in the process of product development, cleaning process is designed, transferred, scaled-up, and then determined. The detail information for Cleaning process design and development refer to separate documentation TR-1400210310 <LSN4182928(ASYM-140021) Step 1 Cleaning Process Design and Development Report>.

2) Stage 2: Cleaning process validation to ensure that the cleaning process developed and designed can repeatedly produce the desired results. This document Cleaning Validation Protocol CVP-1400210301.01 is intended to demonstrate that the approved cleaning procedures consistently remove process residues from product-contact equipment used in the manufacture of ASYM-140851 Step 1 and maintain equipment in a state suitable for subsequent manufacture.

3) Stage 3: Continued cleaning process verification, to ensure that the key parameters and cleaning effect of the cleaning process are continuously monitored, and the cleaning effect is regularly confirmed to ensure that the cleaning process is in a controlled status. The protocol for Continued cleaning process verification will be separately raised after Cleaning process validation defined in this document during subsequent routine commercial production.

The cleaning process validation execution will be performed from the first commercial manufacturing campaign supporting Process Performance Qualification activities. Cleaning validation data will be generated from three consecutive cleaning cycles performed under routine operating conditions by trained production personnel using approved manufacturing and cleaning procedures. Successful completion of three consecutive cleaning validation runs without unexplained failures, deviations impacting cleaning effectiveness, or out-of-limit analytical results will constitute successful cleaning process qualification. For cleaning validation strategy diagram within Asymchem, see Fig. 2.

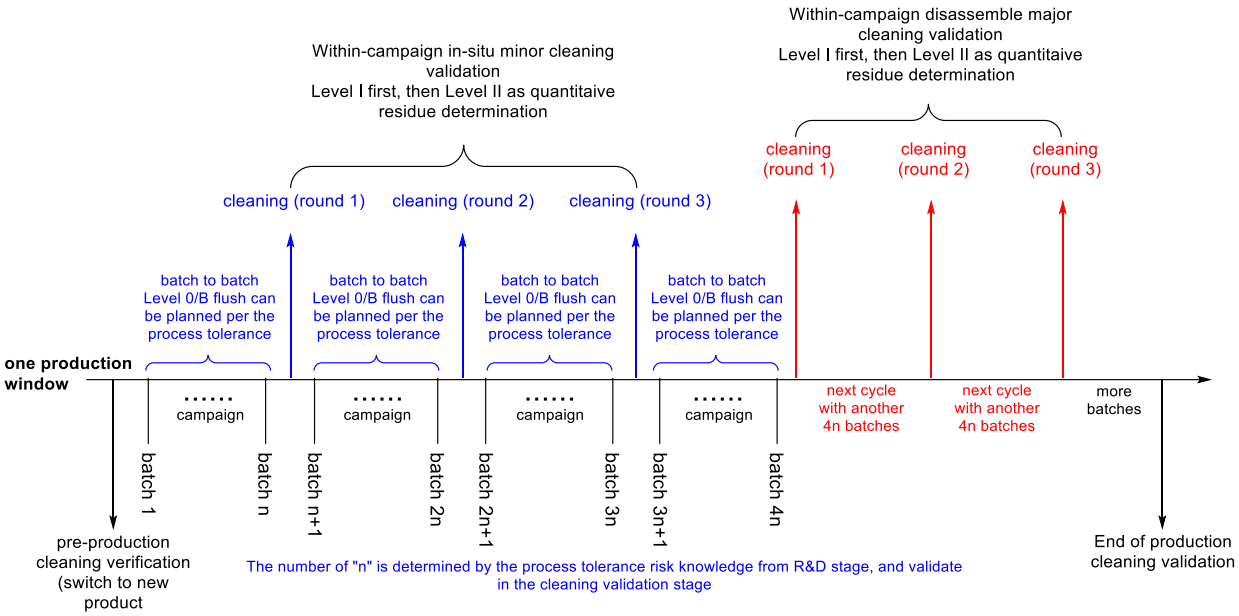


Fig. 2. General cleaning validation strategy

For definition of cleaning criteria:

The cleaning criteria for equipment at each level have been defined in DS-GMP-003 <Equipment cleaning management procedure for GMP productions (DS)>. There are three levels of cleaning check, including Level 0, Level I, and Level II.

Level 0 is further classified into two scenarios: Level 0/A and Level 0/B (batch-to-batch flush).

- **Level 0/A**: The equipment can be successively used between batches without any treatment.
- **Level 0/B** (batch-to-batch flush): The equipment will be simply flushed in-between batches prior to use in the next batch per the process tolerance requirement.

Level I: Visual check on the inner surface of the equipment body after cleaning to confirm it is free of any material residue under adequate light. Equipment cleaning is performed in accordance with Equipment Cleaning Method CMb-TJ414002101 and CMa-TJ414002101. Level I visual inspection shall be carried out on equipment under dry condition upon completion of final nitrogen drying. Lighting conditions and inspection aids are specified, and designated inspection points are defined. All cleaning and inspection activities are documented in the cleaning records, which capture lighting conditions, inspection aids, designated inspection points and equipment components. The inspection results are signed off by the QA personnel.

Level II: Quantitative determination of the target residue on the equipment surface to ensure carryover of the target residue NMT Allowable/Acceptable Residue Limit (ARL) calculated based on health data. Results shall be presented in numerical form, rather than relying solely on a Pass/Fail assessment.

For cleaning frequency

1) In-situ minor cleaning

Within-campaign in-situ minor cleaning: To avoid the following batches being contaminated with residues and degradants from the previous batch, the equipment will be in-situ cleaned (un-disassemble) at defined batch frequency. This validation includes 3 rounds of cleaning and quantitative residue testing (Level I check followed by Level II quantitative test). After completion of the 3 rounds of cleaning validation, the equipment shall be cleaned per the validated cleaning method and batch frequency without extra residue testing.

Based on the cleaning frequency assessment, within campaign in-situ minor cleaning is not involved during manufacture of ASYM-140851 Step 1. The solid-phase synthesis reactor has a complex internal structure, and peptide resin exhibits poor solubility. So, Solid Phase Synthesizer (TJ4S-1213-Sy16) will be disassemble major cleaned after every batch, instead of within campaign in-situ minor cleaning.

Batch to batch flush: Equipment with higher risk will be flushed after each batch as per Level 0/B procedure. Based on the cleaning frequency assessment, batch to batch flush will be performed for key equipment DIC/DMF solution preparation tank (TJ4S-1200-AT01) and Iodine/DMF solution tank (TJ4S-1213-AT09). The DIC/DMF solution preparation tank (TJ4S-1200-AT01) and Iodine/DMF solution tank (TJ4S-1213-AT09) will be batch to batch flushed after every batch.

2) Disassemble Major cleaning：

- **Pre-production cleaning:** To ensure that there is no cross contamination from previous production, and before starting the production activity, all the equipment shall be cleaned to meet the defined ARL (Allowable Residual Limit) or MACO (Maximum Allowable Carry Over) in terms of the previous product in each piece of equipment of the equipment train.
- **Within Campaign cleaning:** To further eliminate the possible impact to subsequent batches from the traceable residual after designed in-situ minor cleaning (un-disassembled), the equipment will be disassembled cleaned at defined batch frequency. Considering the in-situ minor cleaning is validated in three cycles, after run the forth cycle, the next in-situ cleaning may be at risk, so the within campaign disassemble major cleaning is planned at this point to refresh the equipment train's cleanliness. Hence the maximum continuous batches before disassemble major cleaning is defined as: 4n (n stands for the maximum batches for in-situ minor cleaning).

  Based on the cleaning frequency assessment, disassemble major cleaning will be performed after each batch, so the 4n in-situ minor cleaning frequency is not applicable. The solid-phase synthesis reactor has a complex internal structure, and peptide resin exhibits poor solubility. So, Solid Phase Synthesizer (TJ4S-1213-Sy16) will be disassemble major cleaned after every batch.
- **End of production cleaning**: In order to prevent the next product from being contaminated from residue of this product remaining on the equipment, at the end of the production, the equipment will be cleaned following the post-production cleaning method—disassemble cleaning. Disassemble cleaning method will be verified by testing the residue in each piece of equipment separately by sampling from the worst-case location. Three round of cleaning verification consists of cleaning validation.

The disassemble cleaning end of production follows the same implementation method as the disassemble cleaning within campaign, with identical acceptance criteria. Anyway, after campaign the Solid Phase Synthesizer (TJ4S-1213-Sy16) could be disassemble cleaned directly with the validated disassemble cleaning method. For product changeover, the Solid Phase Synthesizer (TJ4S-1213-Sy16) is subject to disassembly major cleaning per CMa-TJ414002101. The ARL will be calculated based on health-based data of the subsequent product, and target residues will be sampled and tested.

According to above cleaning validation strategy (including cleaning criteria and cleaning frequency), for the Step 1 manufacture of ASYM-140021, the cleaning frequency and criteria for each key equipment have been evaluated based on existing data and historical production experience, the detail risk assessment refer to the specific part within this cleaning validation protocol, and the brief information has been summarized as follow in **Table 4**:

- The DIC/DMF solution preparation tank (TJ4S-1200-AT01) and Iodine/DMF solution tank (TJ4S-1213-AT09) will be batch to batch flushed after each batch as per Level 0/B criteria. They were assessed as having no target residue (more detail see the Section 6.5), hence, chemical residue testing is not required, only perform the Level 0/B verification.
- Solid Phase Synthesizer (TJ4S-1213-Sy16) will be disassemble major cleaned after every batch. Therefore, it does not involve in-situ minor cleaning.

Table 4, The basic cleaning information for the equipment

Equipment name	Equipment ID	Cleaning frequency	Cleaning effectiveness verification	Residue limits and basis
Solid Phase Peptide Synthesizer	TJ4S-1213-Sy16 TJ4S-1213-Sy16-R01	Disassemble cleaning after every batch	1. Level I visual check after disassemble cleaning 2. Level II chemical residue test after disassemble cleaning, including Rinse sample and Swab sample	The lower between 10ppm and the calculated allowable residue limit (ARL) based on the health data of compound
Agitation tank	TJ4S-1213-AT09	Batch to batch flushing after each batch	Level 0/B visual check after flush by pre-defined solvent	Only solvent is used, no need for chemical residue test.
		Disassemble cleaning after end of production	Level I visual check after disassemble cleaning	
Agitation tank	TJ4S-1200-AT01	Batch to batch flushing after each batch	Level 0/B visual check after flush by pre-defined solvent	Only solvent is used, no need for chemical residue test.
		Disassemble cleaning after end of production	Level I visual check after disassemble cleaning	

In addition to consider the production arrangement, total 11 continuous batches are planned in the first commercial manufacturing campaign supporting Process Performance Qualification activities, including 3 PPQ batches (CPo140021-03-01-02, CPo140021-03-01-03, CPo140021-03-01-04) and 8 commercial batches starting from October 2026. Hence, the scope of this cleaning validation has been specific determined, refer to **Fig. 3**.

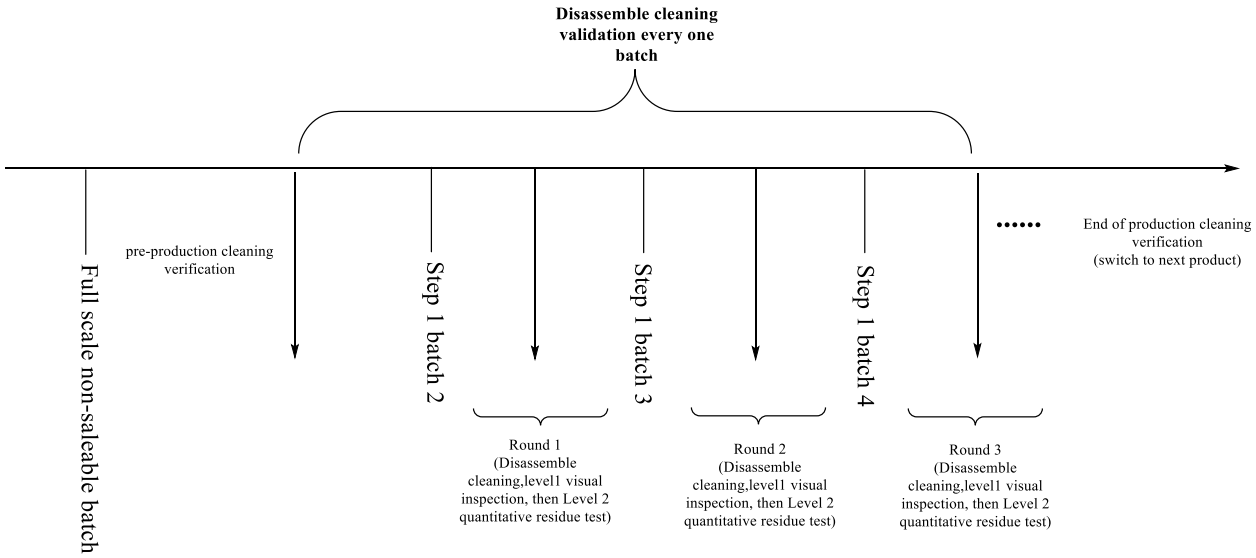


Fig. 3, Solid Phase Synthesizer (TJ4S-1213-Sy16) cleaning validation strategy

Any changes in the cleaning approach post-validation, will be assessed for any impact on the validated cleaning method, and if needed, the cleaning method will be re-validated.

6. Execution procedure

6.1. Summary of equipment train used in the production

The equipment train is intended for CPo140021-03 Step 1 manufacturing, is listed in **Table 5**.

Table 5, Intended equipment train in CPo140021-03 Step1

Step No.	Equipment name	Equipment ID	Size	MOC	Function
1	Peptide Solid Phase Synthesizer	TJ4S-1213-Sy16	3000 L	Hastelloy	AAs activation, Peptide synthesis, Fmoc deprotection, Wash resin, Drying
	Activation tank[2]	TJ4S-1213-Sy16-R01	1500L	SS316	
	Agitation tank	TJ4S-1200-AT01	500 L	Hastelloy	DIC/DMF solution preparation
	Agitation tank	TJ4S-1213-AT09	500 L	Glass-lined	Iodine/DMF solution preparation

Note2: *The Activation Tank (TJ4S-1213-Sy16-R01) is part of the Peptide Solid Phase Synthesizer (TJ4S-1213-Sy16) and is cleaned together with the entire unit. This document will discuss the complete Peptide Solid Phase Synthesizer (TJ4S-1213-Sy16).*

Table 6, Auxiliary equipment in CPo140021-03 Step1

Equipment name(ID)	Auxiliary equipment	Function	Cleaning Cycle
Peptide Solid Phase Synthesizer (TJ4S-1213-Sy16)	Solid sampler (TJ4S-1213-Sy16-Sam01)	Sampling	Perform Level 0/B cleaning after each sampling within batch. Cleaning after production or between batches with Peptide Solid Phase Synthesizer (TJ4S-1213-Sy16)

Equipment name(ID)	Auxiliary equipment	Function	Cleaning Cycle
	Receiving Rake (TJ4S-1213-Sy16-Ra01)	Receiving	Cleaning after production or between batches with Peptide Solid Phase Synthesizer (TJ4S-1213-Sy16)
	In-line Sampler	Sampling	
	Glove box	Receiving	
Activation tank (TJ4S-1213-Sy16-R01)	Metering pump (TJ4S-1213-Sy16-P04)	Add DIPEA	Cleaning after production or between batches with Peptide Solid Phase Synthesizer (TJ4S-1213-Sy16)
	Metering pump (TJ4S-1213-Sy16-P03)	Add DIC	
	System Pump (TJ4S-1213-Sy16-P02)	From TJ4S-1213-Sy16-R01 to TJ4S-1213-Sy16	
Agitation tank (TJ4S-1200-AT01)	Diaphragm Pump (TJ4S-1200-AT01-P01)	Add DIC/DMF Solution	Perform Level 0/B cleaning between batches with Agitation tank (TJ4S-1200-AT01) Perform Level I Cleaning after production with Agitation tank (TJ4S-1200-AT01)
	Mass Flow Meter		
Agitation tank (TJ4S-1213-AT09)	Diaphragm Pump (TJ4S-1107-MPu003)	Add Iodine/DMF solution	Perform Level 0/B cleaning between batches with Agitation tank (TJ4S-1213-AT09) Perform Level I Cleaning after production
	Tantalum mass Flow Meter		
Pipeline	140021-step01-P02	From TJ4S-1213-AT09 to TJ4S-1213-Sy16	Perform Level 0/B cleaning between batches with TJ4S-1213-AT09 Perform Level I Cleaning separately after production
	140021-step01-P01	From TJ4S-1200-AT01 to TJ4S-1213-Sy16	Perform Level 0/B cleaning between batches with TJ4S-1200-AT01 Perform Level I Cleaning separately after production

6.2. Pre-production equipment cleaning check

Before using the equipment for manufacture, all the equipment shall be cleaned. The details of previous product produced in each piece of equipment in the planned equipment train, the calculated MACO (Maximum Allowable Carry Over) of the target residue compound (absolute data), and the corresponding ARL (Allowable Residual Limit, relative data which is calculated from MACO and minimum batch size of next product) are listed in **Table 7**. The relative limit of target residue compound shall be either NMT 10 ppm or the calculated ARL (Allowable Residual Limit), whichever is lower.

Within the campaign (ASYM-140851 into ASYM-140851), the quality-based limit of 10ppm is applicable, and a health-based limit is not needed.

For the Agitation tank TJ4S-1213-AT09, which is used for Iodine/DMF solution preparation, must pass the starch-potassium iodide test paper test before production, and the acceptance criterion is that the starch-potassium iodide test paper shows no color change.

Table 7, Calculated MACO and ARL

Equipment ID	Prev. Product	Prev. product health data/	Next product/	Next product MTDD	MACO of prev. product	Next product min. batch size)	Calculated AR	Calculated ARL, compared to 10 ppm, which is lower
TJ4S-1213-Sy16	ASYM-140851	OEL:10-100ug/m^3	ASYM-140851	$MTDD_{next}$ =1000mg	1576.9mg	157.69kg	ARL=PDE/ $MTDD_{next}$ =OEL×V/$MTDD_{next}$ =10μg/m^3×10m^3/day/1000mg/day =100ppm>10ppm	NMT 10ppm
TJ4S-1200-AT01	None	None	ASYM-140851	None	None	157.69kg	None	None
TJ4S-1213-AT09	None	None	ASYM-140851	None	None	157.69kg	None	None

Note3: *TJ4S-1200-AT01 is used exclusively for DIC/DMF solution preparation, with extremely low residual risk and therefore no target substances.TJ4S-1213-AT09 is used exclusively for iodine/DMF solution preparation, with extremely low residual risk and therefore no target substances.*

The calculation of MACO and ARL following the formula as specified in DS-GMP-003 < Equipment cleaning management procedure for GMP production> (current version), for example:

$$ARL = \frac{PDE}{MTDDnext}$$

$$MACO = ARL \times MBS = \frac{PDE \times MBS}{MTDDnext}$$

The acronyms in the above equations are listed in **Annex 5**. The release result of the equipment train will summarize into <Equipment Training release checklist> (QF03282)

6.3. Batch to batch flushing to mitigate the risk in the high risk equipment

The equipment with high risk shall be cleaned after each batch per Level 0/B criteria to mitigate the risk per step1 process tolerance, as summarized in **Table 8**. According to the assessment in **Table 8**, the equipment TJ4S-1200-AT01 and equipment TJ4S-1213-AT09 show high risks, respectively, thus they will be subjected to batch-to-batch flushing. The batch-to-batch flushing cleaning method will shall be defined in CMb-TJ414002101 (current version).

Table 8, Risk assessment on batch to batch flushing

Step No.	Equipment name	Equipment ID	Function	Risk assessment	Conclusion
1	Peptide Solid Phase Synthesizer	TJ4S-1213-Sy16 (3000L)	Amino acid activation, peptide synthesis, Fmoc deprotection, Wash resin, Drying	This equipment is critical for solid-phase synthesis. The solid-phase synthesis reactor has a complex internal structure, and peptidyl resin exhibits poor solubility, making it hard to be cleaned completely by batch to batch flush. Inadequate cleaning will directly compromise the product quality of subsequent batches. In the solid-phase synthesis reactor, 2-CTC resin undergoes repeated cycles of coupling, washing, de-protection, re-washing, I_2 oxidation, last drying to produce the	Disassembled major cleaning is performed for every batch; therefore, batch-to-

Step No.	Equipment name	Equipment ID	Function	Risk assessment	Conclusion
				peptidyl resin in-process intermediate ASYM-140851. Incomplete cleaning within campaign may lead to formation of additional long-chain impurities during the reaction cycles of the next batch. The reactor shall be disassembled for thorough cleaning upon completion of each batch to ensure no solid peptide resin residues remain, performing disassemble major cleaning every batch, so no need batch to batch flush.	batch flush is not required.
2	Agitation tank	TJ4S-1200-AT01	Prepare DIC /DMF solution	This tank is dedicated to the preparation of DIC/DMF solution, where DIC and DMF are charged accordingly. DIC is chemically compatible with anhydrous DMF; its primary degradation pathway is hydrolysis (product: DIU, no coupling activity). Fresh solution is generally prepared for immediate use with short-term storage at 10–30 °C. The equipment remains closed and maintained under nitrogen blanket between batches. During continuous production, the residual DIC in the reactor may lead to the accumulation of impurities. DIC is easily removed by using DMF through batch to batch flushing. Based on above, it is low risk to perform batch to batch flushing. Therefore, after each batch is completed, the Agitation tank needs to conduct batch to batch flush by spray ball (coverage test qualification refer to SVP-TJ4-170.01-067).	After each batch is completed, the Agitation tank needs to conduct batch to batch flush.
3	Agitation tank	TJ4S-1213-AT09	Prepare Iodine /DMF solution	This tank is dedicated to iodine/DMF solution preparation for solid-phase synthesis, with nitrogen blanketing between batches. Iodine in DMF may weakly interact with DMF under prolonged heating or storage. Residual iodine from the previous batch (partially degraded) and its degradation products (such as iodide ions, DMF-iodine complexes, etc.) can carry over to the next batch. Without cleaning between batches, impurity levels accumulate progressively over multiple batches. Iodine is easily removed by using DMF batch to batch flushing. Therefore, after each batch is completed, the Agitation tank needs to conduct batch to batch flush by spray ball (coverage test qualification refer to SVP-TJ4-170.01-064).	After each batch is completed, the Agitation tank needs to conduct batch to batch flush

6.4. Selection of the target residue compounds

As per the process of Step 1, the following materials are selected as the target residue compound for testing, see **Table 9**. And the rationale for target residue compound selection of each equipment is listed in **Table 10**.

Table 9, Target residue compound selected for testing

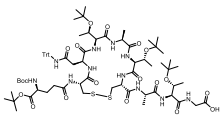
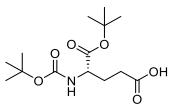
Target Compound	Asymchem code	Customer code	Structure	Batch size/kg	Solubility data
N-((((4R,7S,10S,13S,16S,19R)-19-((S)-5-(tert-butoxy)-4-((tert-butoxycarbonyl)amino)-5-oxopentanamido)-7,13-bis((R)-1-(tert-butoxy)ethyl)-10-methyl-6,9,12,15,18-pentaoxo-16-(2-oxo-2-(tritylamino)ethyl)-1,2-dithia-5,8,11,14,17-pentaazacycloicosane-4-carbonyl)-L-alanyl)-O-(tert-butyl)-L-threonylglycine	ASYM-140852	LSN4214882		157.69~292.80	~1 g in 20 ml DMF at 20 - 25 ℃
Boc-L-Glu-OtBu	ASYM-140413	N/A		157.69~292.80	~1 g in 5 ml DMF at 20 - 25 ℃

Table 10, Rationale of target residue compound selection

Step No.	Equipment name	Equipment ID	Function	Target residue	Rationale of target residue compound selection
1	Peptide Solid Phase Synthesizer	TJ4S-1213-Sy16 (3000L)	Peptide synthesis, Fmoc deprotection, Wash resin, Drying	ASYM-140852 and Boc-L-Glu-OtBu	Combining the process characteristics of the solid-phase synthesis reactor (including synthesizer and activation tank) and the subsequent soft cleavage process route of ASYM-140851 peptidyl resin, the peptide fragments bound to CTC resin are easily dissociated from the resin carrier in advance, forming the free solid product ASYM-140852, which had been proved by shaking the peptidyl resin ASYM-140851 in DMF solvent, it was demonstrated that the filtrate was mainly detected of the cleavage product ASYM-140852. The ASYM-140852 becomes the main pollutant on the equipment surface. ASYM-140851 peptidyl resin is a resin-bonded product, and the resin particles are easy to remove by washing; ASYM-140852 exists as fine solid particles, which are more liable to adsorb onto equipment surfaces and present greater cleaning difficulty. Material drying is the final unit operation carried out in the solid-phase synthesis reactor. No additional solvents are introduced in this step, and the process intermediate ASYM-140852 demonstrates favorable stability under drying conditions, such that residues on equipment surfaces are not prone to spontaneous degradation or elimination. In consideration of all the above factors, Process

					intermediate ASYM-140852 is specified as the target residue for cleaning validation of this equipment.
					The activation tank is exclusively used for amino acid activation. The activated amino acid solution is transferred to the next synthesizer unit via a dedicated pipeline, followed by washing and the next amino acid activation cycle. The tank does not come into contact with resin throughout the entire process. A total of 6 amino acids are used in this equipment. After the activation and transfer of each amino acid is completed, the tank is washed three times with DMF, which has good solubility for amino acids. The peptide solid-phase synthesizer is cleaned as a whole. Boc-L-Glu-OtBu was additionally selected as another target residue because it represents the final activated amino acid processed within the activation tank immediately prior to completion of the synthesis sequence. Due to its repeated exposure to activation and transfer operations and its potential presence within transfer lines and activation system components, it represents an additional worst-case cleaning challenge. Boc-L-Glu-OtBu is stable in the cleaning solvent. Therefore, Boc-L-Glu-OtBu and ASYM-140852 are selected as target residues.
2	Hastelloy agitation tank	TJ4S-1200-AT01 (500L)	DIC / DMF solution preparation	N/A	TJ4S-1200-AT01 is exclusively used for the preparation of DIC/DMF solution. DIC is a small-molecule coupling reagent, not an active pharmaceutical ingredient (API) or key process intermediate. It exhibits favorable solubility in the selected cleaning solvent (DMF) and can be readily removed by rinsing. The equipment is kept in closed state under nitrogen atmosphere in-between batches. Therefore, it is not necessary to quantitatively test the target residue.
3	Glass-lined agitation tank	TJ4S-1213-AT09 (500L)	Iodine /DMF solution solution preparation	N/A	TJ4S-1213-AT09 is exclusively used for storage of iodine/DMF solution. Iodine is a small-molecule inorganic reagent, not an active pharmaceutical ingredient (API) or key process intermediate. It exhibits favorable solubility in the selected cleaning solvent (DMF). This equipment only contacts homogeneous solution with low cleaning difficulty. The equipment is kept in closed state under nitrogen atmosphere in-between batches. Therefore, it is not necessary to quantitatively test the target residue.

6.5. Cleaning frequency

6.5.1 Cleaning frequency for within campaign in-situ minor cleaning

The equipment cleaning frequency and justification is further indicated in **Table 11**. The cleaning frequency has been established based on process knowledge, equipment design, residue characteristics and manufacturing experience.

Table 11, Rationale of the equipment batch cleaning frequency

Step No.	Equipment name	Equipment ID	Function	Target residue	Cleaning Frequency	Rationale of Cleaning Frequency
1	Peptide Solid Phase Synthesizer	TJ4S-1213-Sy16 (3000L)	Peptide synthesis, Fmoc de-protection, Wash resin, Drying	ASYM-140852	Disassemble Major cleaning every batch	This equipment is critical for solid-phase synthesis. The solid-phase synthesis reactor has a complex internal structure, and peptide resin exhibits poor solubility, making it hard to rinse completely. Inadequate cleaning will directly compromise the product quality of subsequent batches. In the solid-phase synthesis reactor, 2-CTC resin undergoes repeated cycles of coupling, washing, de-protection and re-washing to produce the peptidyl resin in-process intermediate ASYM-140851. Incomplete inter-batch cleaning will lead to formation of additional long-chain impurities during the reaction cycles of the next batch. Based on past production experience of full-scale non-saleable batches, the reactor shall be disassembled for thorough cleaning upon completion of each batch to ensure no solid peptide resin residues remain. Due to performing disassemble major cleaning every batch, so no need in-situ minor cleaning.
	Agitation tank	TJ4S-1200-AT01	DIC/DMF solution preparation	N/A	Batch-to-batch flush (Level 0/B); no in-situ minor cleaning; disassemble major cleaning at end of production	Per CPo140021-02 (RS campaign), the DIC/DMF solution is stable for 2 days at 10–30 °C. The agitation tank is maintained under nitrogen blanket, so no water intrusion will occur. During continuous production, after the process mixture in this reactor is fully transferred downstream, the next batch is charged into the reactor. All residual material in the tank will be fully dissolved and carried over into the next batch, with no residual accumulation across multiple batches. DIC is fully miscible with DMF at any ratio, and batch to batch flushing is performed with DMF. Based on the above risk

Step No.	Equipment name	Equipment ID	Function	Target residue	Cleaning Frequency	Rationale of Cleaning Frequency
						assessment, it is low risk to perform the flushing every batch. The cleaning frequency is defined as formal cleaning after completion of the entire production campaign. Therefore, only batch to batch flush is required; in-situ minor cleaning is not necessary.
	Agitation tank	TJ4S-1213-AT09	Iodine / DMF solution preparation	N/A	Batch-to-batch flush (Level 0/B); no in-situ minor cleaning; disassemble major cleaning at end of production	According to CPo140021-02 (RS campaign), residual solution from the previous batch stored for up to 2 months has no impact on the composition and impurity profile of the next batch. The iodine/DMF solution is stable at 10~30 °C. During continuous production, after the process solution in this reactor is fully transferred downstream, the next batch is charged into the reactor. All residual material in the reactor will be fully dissolved and carried over into the next batch, with no residual accumulation across multiple consecutive batches. Iodine has favorable solubility in DMF. Based on the above risk assessment, it is low risk to perform the flushing every batch. The cleaning frequency is defined as formal cleaning upon completion of the entire production campaign. Therefore, only batch to batch flush is required; in-situ minor cleaning is not necessary.

6.5.2 Cleaning frequency for dis-assemble major cleaning

Dis-assemble major cleaning shall be carried out after each batch manufacturing completion and switching to next product. The disassemble major cleaning method for end of production is same with disassemble major cleaning method for within campaign, hence dis-assemble major cleaning will be directly performed after each campaign once the within campaign dis-assemble major cleaning validation completion.

6.6. Allowable residue limit determination

6.6.1 Allowable residue limit for within-campaign in-situ minor cleaning

Based on above risk assessment about the within-campaign cleaning frequency, within-campaign in-situ cleaning isn’t applicable, hence this part isn’t applicable within this cleaning validation protocol.

6.6.2 Allowable residue limit for dis-assemble major cleaning

After production of one batch has ended, the equipment shall be cleaned following the end of production cleaning method: disassemble cleaning, all the equipment shall be quantitatively tested to verify as meeting the defined ARL (or MACO) in terms of the previous product in each piece of equipment, the target compound relative residue criteria shall be the stricter level between 10 ppm (which is Asymchem's internal acceptance criteria defined in DS-GMP-003 <Equipment cleaning management procedure for GMP production> (current version)) and calculated allowable residual limit (ARL). The calculation of MACO and ARL following the formula as specified in DS-GMP-003 <Equipment cleaning management procedure for GMP production> (current version).

Considering the within-campaign disassemble cleaning frequency is defined as cleaned after every batch for peptide solid phase synthesizer, and the target of selected equipment is listed in **Table 12**, and the calculation based on established healthy data is listed in **Table 13**.

Table 12, Toxicological data of the equipment selected for chemical residue test

Step No.	Equipment name	Equipment ID	Function	Target residue	Toxicological data
1	Peptide Solid Phase Synthesizer	TJ4S-1213-Sy16 (3000L)	Peptide synthesis, Fmoc deprotection, Wash resin, Drying	ASYM-140852	OEL:10-100ug/m^3
				Boc-L-Glu-OtBu	OEL:10-100ug/m^3

Table 13, Calculated MACO and ARL

Equipment ID	Prev. Product	Prev. product health data/	Next product/	Next product MTDD	MACO of prev. product	Next product min. batch size)	Calculated AR	Calculated ARL, compared to 10 ppm, which is lower
TJ4S-1213-Sy16	ASYM-140851	OEL:10-100ug/m^3	ASYM-140851	$MTDD_{next}$ =1000mg	1576.9mg	157.69kg	ARL=PDE/ $MTDD_{next}$ =OEL×V/$MTDD_{next}$ =10μg/m^3×10m^3/day/1000mg/day =100ppm>10ppm	NMT 10ppm
	Boc-L-Glu-OtBu	OEL:10-100ug/m^3	ASYM-140851	$MTDD_{next}$ =1000mg	1576.9mg	157.69kg	ARL=PDE/ $MTDD_{next}$ =OEL×V/$MTDD_{next}$ =10μg/m^3×10m^3/day/1000mg/day =100ppm>10ppm	NMT 10ppm

Table 14, The calculated actual swab limit and rinse limit

Step No.	Equipment ID	Target residue (Asymchem compound code)	Equipment material	Equipment internal surface area /Weight range and type of rinsing & refluxing solution	After Campaign			Between Campaign		
					Equipment minimum output (e.g., refer to QM05004)	ARL	ARL of per cm^2/100cm^2 area/per unit volume	Minimum batch size of next product	ARL	ARL of per cm^2/100cm^2 area/per unit volume
Step 1	3000L TJ4S-1213-Sy16	140852	Hastelloy	☒Swab：208185cm^2 ☐Reflux：N/A ☒Rinse：DMF 105~115kg (111~121L)	7.5 kg	10	Swab：36.03μg/100cm^2 Rinse：0.00062~0.00067mg/mL	157.69kg	10	Swab：757.45μg/100cm^2 Rinse：0.0129~0.0142mg/mL
		140413 (YL008618)	Hastelloy	☒Swab：208185cm^2 ☐Reflux：N/A ☒Rinse：DMF 105~115kg (111~121L)	7.5 kg	10	Swab：36.03μg/100cm^2 Rinse：0.00062~0.00067mg/mL	157.69kg	10	Swab：758.94μg/100cm^2 Rinse：0.0129~0.0142mg/mL

6.6.3Rationale for Exemption of Microbial/Bioburden Assessment for SPPS

Bioburden/microbial and endotoxin contamination were assessed as low risk Through evaluation of the anhydrous synthesis process, equipment washing with water followed by rinsing with DMF/MTBE and nitrogen drying, as well as downstream processes and control measures. The detailed assessment refers to the document RA-140021030103 < Risk Assessment Report for Contamination Control Strategy in LSN4182928/ASYM-140021 production>.

6.7. Equipment cleaning evaluation based on the risk analysis

- **Assessment of the equipment**

As per the residue material's solubility, each equipment has a separately designed cleaning method, see **Annex 1**. All the equipment will be cleaned in a defined interval according to cleaning method < CMb-TJ414002101 (current version)>. Not all the equipment will be swabbed, rinsed or refluxed to sample for residue test. Only the equipment that is confirmed as higher contamination risk will be swabbed, rinsed or refluxed to sample for the residue test after cleaning.

The risk assessment is based on the Risk Ranking and Filtering method according to ICH Q9<Quality Risk Management>, ASM-027 <Application of risk management tools> and QAP-005 <Risk Management Policy>. The selected method to determine worst case equipment -- risk ranking and filtering method refer to Asymchem's SOP <Cleaning Validation> DS-GMP-043 (current version), detail assessment of worst case equipment is shown as follow:

The standard of the assessment is set as indicated in **Table 15**, and the standard risk grade is laid down in the four typical aspects with regard to the direct product contact equipment: material of construction, smoothness of the surface, complexity of the equipment construction, function of the equipment. In each of the four aspects, the risk grade is set in different circumstances. Basically, the parameters of GL reactor with homogenous reaction are set as 1.

Table 15, Standard risk ranking in the risk assessment

Material of construction of the direct product contact (A)	**Stainless steel(SS), Hastelloy**	**Glass-lined(GL)**	**PTFE, PE, Halar**	**Polypropylene(PP)**
Risk ranking by score	0.8	1	2	3
Smoothness of the surface (direct product contact) (B)	Flat smooth	Curve smooth	Irregular geometry smooth	Rough
Risk ranking by score	0.5	1	2	3
Complexity of the equipment construction (C)	Single geometry space	2 or 3 continuous geometry space	Muti-continuous geometry space	Complex irregular geometry space
Risk ranking by score	1	1.5	2	3

Function of the equipment（D）	Prepare inorganic solution	Prepare organic solution	Homo-genous workup	Homo-genous reaction	Hetero-geneous workup	Hetero-geneous reaction	Centrifuge or filter	Only solid material contact (drying or material charging)	Crystallization, recrystallization
Risk ranking by score	0.5	0.6	0.8	1	1.5	1.8	2.5	2.8	3

Based on the above established standard, risk assessment is performed on all the equipment used in the process. The result is listed in **Table 16**. The risk score is calculated by multiplying the grade score of A, B, C and D. And A stands for grade score of Material of construction of the direct product contact, B stands for Smoothness of the direct product contact surface, C stands for Complexity of the equipment construction, D stands for Function of the equipment. A**f**ter obtaining the risk score (gray background), the equipment with the same residue target, the same cleaning frequency, is kept under one group for comparison, and the highest score equipment shall be selected for the residue test as worst case. After risk filtering in Section 6.5, the Solid Phase Synthesizer (TJ4S-1213-Sy16) is cleaned by disassembly after each batch with highest score, while the agitation tank (TJ4S-1200-AT01) and agitation tank (TJ4S-1213-AT09) do not involve chemical residue testing. Hence, Peptide Solid Phase Synthesizer TJ4S-1213-Sy16 is selected as the typical and representative equipment for chemical residue test.

Table 16, Risk Assessment Based on Risk Ranking and Filtering method

Target residue	**Cleaning frequency**	**Equipment type**	**Equipment ID**	**Size**	**MOC**	**Function**	**Risk Assessment (**in terms of direct product contact**)**					**Risk filtering**
							Material of construction	Smoothness of the surface	Complexity of the equipment construction	Function of the equipment	**Risk score ***	
							A	**B**	**C**	**D**		
ASYM-140852	Every one batch	Peptide Solid Phase Synthesizer	TJ4S-1213-Sy16	3000 L	Hastelloy	Peptide synthesis, Fmoc deprotection, Wash resin, Drying/	0.8	2	2	2.8	8.96	Typical and Representative
N/A	Every one batch	agitation tank	TJ4S-1200-AT01	500L	Hastelloy	DIC/DMF solution preparation	0.8	1	1	0.5	0.4	Not Typical and Representative (Not involve chemical residue testing)
N/A	Every one batch	agitation tank	TJ4S-1213-AT09	500L	Glass-lined	Iodine / DMF solution preparation	1	1	1	0.5	0.5	Not Typical and Representative (Not involve

Target residue	Cleaning frequency	Equipment type	Equipment ID	Size	MOC	Function	Risk Assessment (in terms of direct product contact)					Risk filtering
							Material of construction	Smoothness of the surface	Complexity of the equipment construction	Function of the equipment	Risk score *	
							A	B	C	D		
												chemical residue testing)

** Risk score=A×B×C×D*

‡ Direct material contact, and material is easy to stick on the equipment, or the equipment is of complex structure is evaluated as "higher risk" in the risk ranking

† If one equipment experienced multi-purpose in the process, the highest risk score shall be considered

§In terms of the same residue target equipment, if final risk score is the same, the more complex structured equipment shall be considered as the typical and representative; and if two equal equipment were included for the same process (for example, two SS centrifuge machines are used for the isolation of the same product), either one can be considered typical and representative.

*After the above risk ranking and filtering, the following equipment in **Table 17**, assessed as typical and representative equipment shall be rinse sampled or swab sampled for testing after equipment cleaning.*

Table 17, Typical and representative equipment after Risk Ranking and Filtering

Target residue	Cleaning frequency	Equipment type	Equipment ID	Size/Inner surface area	MOC	Function
ASYM-140852	Dis-assemble major cleaning every batch	Peptide Solid Phase Synthesizer	TJ4S-1213-Sy16	3000 L	Hastelloy	Peptide synthesis, Activate, Drying

The Peptide Solid Phase Synthesizer (TJ4S-1213-Sy16) has been identified as the worst-case equipment based on the following: it is in direct contact with peptide-bearing resin throughout the entire synthesis process, representing the longest product contact duration; the equipment has complex geometry with internal features that limit cleaning accessibility; internal structures (such as sieve plates, dead legs and transfer pathways) have a high residue retention potential; and the equipment is exposed to the highest quantity of process-related residues. All of the above factors have been considered, and this equipment received the highest overall score.

- **Assessment of the spare parts**

During the cleaning, the equipment parts such as gaskets, valves, tees, cross, sight glass_U-bend, condenser and reducers, etc. shall be cleaned together with the corresponding equipment.

For the within-campaign in-situ minor cleaning, at the defined intervals in the production, the equipment including the accessories shall be in-situ cleaned and the residue shall be measured by swab samples or solvent flush / rinse or other method (as defined in the cleaning method), without disassemble, before sampling for testing.

- **Assessment of the pipelines**

For the pipelines connections in the equipment train, it is fixed for the dedicated purpose in the process, there are several scenarios:

For the pipelines used for solvent delivery, it only contacts the solvent without other materials/intermediate/products. The solvents in this process are stable, so there is no solvent decomposition. After each addition, purge with nitrogen to ensure that all solvents are added. When at rest, the pipelines are in a closed state, so the cleaning risk level is low.

The pipelines used for the process solution or slurry transfer are cleaned together with the equipment with solvent rinses / flushes, so the risk level is low in terms of the same step in the same project, since pipelines are dedicated to each commercial project.

The pipelines used for the process solution transfer are cleaned together with the equipment with solvent rinses / flushes, so the risk level is low in terms of the same step in the same project, since pipelines are dedicated to each commercial project. In addition, if the production interval for ASYM-140851 will be long in future, the cleaned pipelines also can be assigned to other projects or disposable, then new pipelines will be engaged in next ASYM 140851 manufacturing.

The flexible hose is disposable for project; it will be discarded after campaign.

- **Assessment of the sampling tools**

The glass sampler is disposable and will not be cleaned.

For sampling of the process system in the peptide solid-phase synthesizer, samples will be collected under vacuum through an in-line sampler (stainless steel sampling line + glass collection bottle). Prior to each sampling, the sampling line needs to be purged with nitrogen to blow the system contents back into the process system, followed by vacuum sampling. The first two collected samples are discarded, and the third collected sample is sent for testing. After each sampling use, the glass collection bottle is not cleaned; a new glass collection bottle is replaced for the next sampling. During disassembly and cleaning, the in-line sampler (stainless steel sampling line) will be cleaned together with the peptide solid-phase synthesizer. Alternatively, sampling of the process system in the peptide solid-phase synthesizer can be performed directly using a disposable PTFE tube and a disposable syringe.

For sampling of solid materials in the solid-phase synthesizer, a stainless steel solid sampler is typically used. The stainless steel sampler shall be cleaned after each sampling operation. During disassembly and cleaning, the stainless

steel solid sampler shall be cleaned together with the peptide solid-phase synthesizer. A detailed risk assessment for the cleaning of sampling tools is presented in **Table 18**.

Table 18, Risk evaluation to sampling tool cleaning

No	Tool	Criticality assessment		Residual possibility assessment			Level of risk	Control measure
		Impact on product quality	Criticality	Surface Assessment: Stainless Steel (SS), Plastic (P), Glass (G),Smooth (SM), Rough (R)	Clean Difficulty	Possible cause		
1	Thief	Direct contact with the material. The residue will bring potential cross-contamination if reused.	Critical	SS, SM	Easy to clean as the surface is smooth	Low	Low	Flush/swab the surface completely, visual Inspection

- **Assessment of the operation aids**

For the process aids, for example, the scoop, rake, pump and secondary protection bucket, the cleaning method is separately provided in the cleaning method document CMa-TJ414002101 <Equipments Cleaning Method> (current version) and CMb-TJ414002101 <Equipments Cleaning Method> (current version). Considering the MOC of these operation aids is SS & GL, smooth surface, simple in the structure, be removable cleaned and direct visual inspection, so the material residue is low risk.

- **Assessment of the filter consumables**

In the manufacturing of ASYM-140851, the gloves, latex tube, PTFE pipe and filter cartridge are used which are directly contact with specific material. From which, the gloves, latex tube and filter cartridge, will be disposed after cleaning together with the relevant equipment (Peptide Solid Phase Synthesizer, reactor, liquid filter).

In the SPPS process, there is 10 μm stainless steel filter before the solvent entering the production building from tank farm. This filter is exclusively used for filtering fresh solvents from the tank farm. Preventive maintenance is performed every 6 months (refer to QM03025<Equipment and facility maintenance manual>), including cleaning and integrity testing. Since it does not contact with peptide-related materials, the possibility of cross-contamination transfer is extremely low, thus the risk level is determined as low. so the risk level is low in the project.

6.8. Cleaning method

The Level 0/B batch to batch flush cleaning is included in CMb-TJ414002101<Equipments Cleaning Method> (current version). The disassemble major cleaning method is included in CMa-TJ414002101< Equipments Cleaning Method> (current version).

The cleaning method for the worst-case equipment, the peptide solid phase synthesizer, have been practiced in full scale non-saleable batch, and achieved the cleaning purpose.

The critical cleaning parameters that need to be controlled and recorded in each validation run are shown in the table below.

Table 19, Cleaning Parameter Collection for Peptide Solid Phase Synthesizer

Equipment type	Equipment ID	Cleaning frequency	Cleaning solvent	Cleaning method	Cleaning temp.	Cleaning solvent qty.	Cleaning time	Stir time	Stir speed	N_2 Flow	Soak times	Repeat times
					Pre-defined	Pre-defined	Pre-defined	Pre-defined	Pre-defined	Pre-defined	Pre-defined	Pre-defined
Peptide solid phase synthesizer	TJ4S-1213-Sy16 TJ4S-1213-Sy16-R01	**Disassemble major cleaning** After every batch	DMF	Automatic cleaning	Undefined	290~310kg	3min	15min	20~30rpm	7~13 m^3/h	NLT 0.5h	2
			Purified water		Undefined	290~310kg	3min	15min	20~30rpm	7~13 m^3/h	Not applicable	3
			Purified water	Rinse	Undefined	105~115kg	Undefined	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
			Purified water	Wipe	Undefined	29.0~31.0kg	Undefined	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
			DMF	Automatic cleaning	Undefined	105~115kg	3min	15min	20~30rpm	7~13 m^3/h	Not applicable	1
			MTBE		Undefined	170~190kg	3min	15min	20~30rpm	7~13 m^3/h	Not applicable	1

Table 20, Summary of Equipment Cleaning Parameters for Agitation Tank

Equipment type	Equipment No.	Cleaning frequency	Cleaning solution	Cleaning method	Cleaning solution quantity
					Preset value
Agitation tank	TJ4S-1200-AT01	**Disassemble major cleaning** After end of production	DMF	Rinse	95~105kg
			Purified water	Rinse	100~110kg
			Purified water	Wipe	20~25kg
		Batch to batch flush After each batch	DMF	Flush	90~110kg
Agitation tank	TJ4S-1213-AT09	**Disassemble major cleaning** After end of production	DMF	Rinse	95~105kg
			Purified water	Rinse	100~110kg
			Purified water	Wipe	20~25kg
		Batch to batch flush After each batch	DMF	Flush	90~110kg

6.9. Sampling plan and method

- **Swab *sampling for cleaning validation***

1) Safety

Wear powder-free gloves and goggles; Extreme care should be taken to prevent swab contamination during handling.

2) Materials/Tools:

a) Solvent: For the target compound, use the solvent prescribed in the specific cleaning method.

b) For flat surfaces, use a frame draw the sampling area.

c) An appropriate cuvette.

d) Swab: Use approved swabs (Supplier: Texwipe; Part number: TX715; material: polyester; Swab head: length 25.7mm, width 13mm, thickness 4.2mm; Swab handle: length 101.8mm, width 5.2mm, thickness 3mm), in this validation Texwipe TX715 polyester swab is selected for active ingredients sampling (**Fig. 4**).



Fig. 4, Texwipe TX715 polyester swab

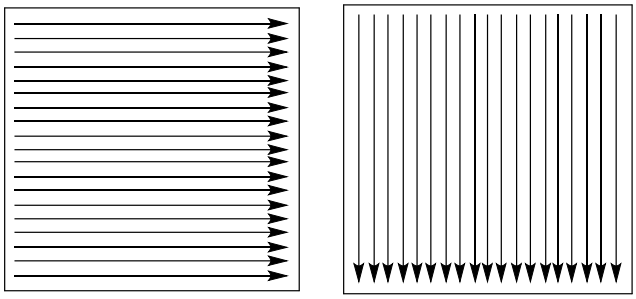


Fig.5 Swab instruction over the sampling surface (10cm*10cm)

3) ***Procedure:***

Press the swab head hard onto the sampling surface at about a 45° angle between the swab and the sampling surface, and make sure that the swab surface is closely in contact with the sampling surface, and then swab the sampling surface slowly and steadily. Use one side of the swab to swab (swab sampling diagram as F**ig. 5**), then turn over the swab and change the swabbing direction to make it perpendicular to the previous swabbing direction (swab sampling diagram as F**ig. 5**). The swabbing process should cover the entire sampling surface. There should be no gaps in the sampling surface that have not been swabbed.

4) ***Swab sampling location determination***

The selected sampling location for each type of equipment was assessed as a general guideline at Asymchem, for detail, see <Equipment to be swab-tested and the sampling points diagram> (QM04005 current edition). The sampling points are selected based on the document RA-TJ42024002.11<Risk Assessment Report for Sampling Point Locations of Production Equipment>, which is based on the risk ranking and filtering method according to ICH Q9, ASM-027<Application of Risk Management Tools> and QAP-005<Risk

Management Policy>. Location selection considered complexity of the equipment construction, MOC of direct product contact part, smoothness of the surface, exposure to peptide-bearing resin and possibility of visual check. The selected locations therefore provide a conservative assessment of overall cleaning effectiveness. The specific sample location during the cleaning validation is summarized in the following **Table 21**.

Table 21, the sample locations for the selected typical equipment

Equipment I.D.	Sample location 1	Sample location 2	Sample location 3	Sample location 4
Peptide Solid Phase Synthesizer (TJ4S-1213-Sy16)	 Agitator blade	 Inner surface	 Side outlet	 Sampling glove box

And further illustrated as PI&D diagram in **Annex 2.**

- **Rinse sampling for cleaning validation**

1) Safety: Wear gloves and goggles
2) Tools: An appropriate sampling bottle
3) Rinse sampling plan: Sampling plan includes sampling location, sample size, test items, sample number, sample method, frequency and testing plan, referring to **Table 22**.
   a) Rinse samples shall be collected only after completion of the equipment cleaning and successful QA visual check inspection.
   b) The sampling solvent will be that which is referred to in the cleaning method document.
   c) Choose the lowest outlet of the equipment to let rinse solvent out.
   d) Gather enough amount from the mid-part of the solution stream from the outlet of the equipment in a sampling bottle.

Table 22, Rinse and Swab sampling plan

Equipment name	Sampling Location	Detail Location Description	Sampling method	Sample size	Sampling position PI&D number	Test item	Number of Sample
Peptide Solid Phase Synthesizer (TJ4S-1213-Sy16)	Upper expansion valve port of reactor	After washing the activation tank, an appropriate amount of eluent in the middle section is received from the upper expansion valve port of the synthesis reactor by running the transfer program and sent for testing	Rinsing sampling	~500ml	Sampling location Diagram for Peptide Solid Synthesize	ASYM-140852 and Boc-L-Glu-OtBu residue	1

Equipment name	Sampling Location	Detail Location Description	Sampling method	Sample size	Sampling position PI&D number	Test item	Number of Sample
	Agitator	Sampling area of inner surface of 100 cm^2 agitator blade (bottom surface centre of agitator blade)	Swab sampling	100_cm^2			1
	Inner-surface of the synthesis reactor	Sampling area of 100 cm^2 inner surface (close to the lower edge of inner surface)	Swab sampling	100_cm^2			1
	Side discharging outlet	Sampling area of inner surface of 100 cm^2 side outlet (close to the outer edge of the side outlet	Swab sampling	100_cm^2			1
	Sampling glove box	Sampling area of inner surface of 100 cm^2 (between left and right edges of the chamber Bottom)	Swab sampling	100_cm^2			1

6.10.Test method validation and recovery study

The cleaning validation studies are based on the swabbing technique and refluxing/rinse technique. The validation studies include swabbing and refluxing of the main surface area as well as the hard-to-clean locations, both of which are carried out for the target compounds.

The validation results of system suitability, specificity, linearity, accuracy, limit of quantification and solution stability of method AM-140021-027.01 and AM-140021-026 all met the acceptance criteria as specified in validation protocol AM-140021-027-VP-01.01(Validation report: AM-140021-027-VR-01) and AM-140021-026-VP-01 (Validation report: AM-140021-026-VR-01), and the maximum injections between the two bracketing standard is 6. The limit of quantification solution, reference standard solution and swab sample solution were stable at room temperature for 24h. The method AM-140021-027.01 and AM-140021-026-VP-01 have been successfully validated by QC laboratory of Asymchem Life Science (TJ4 site), and could meet the detection requirements.

After wipe sampling, seal the sampling bottle and place it in the pending-testing area. After the sample solution has been tested, seal it and place it at room temperature in the pending-disposal area.

Testing methods for target compounds are listed in **Table 23**.

Table 23, Test method of the corresponding target in the cleaning validation

No.	Target	Test method	LOQ for Swab & Rinse	Rinse Limit	Swab Limit
1	ASYM-140852	AM-140021-027	0.00015 mg/mL	0.0129~0.0142 mg/mL	757.45μg /100cm^2
2	Boc-L-Glu-OtBu (YL008618/ASYM-140413)	AM-140021-026	0.0013 mg/mL	0.0129~0.0142 mg/mL	757.45μg/100cm^2

All the test methods shall be validated before performing the cleaning validation. The information of method validation is listed in **Table 24**.

Table 24, Information of method validation

Method Number	Method transfer/validation protocol no.	Method transfer/validation report no.	Qualified Lab
AM-140021-027	AM-140021-027-VP-01	AM-140021-027-VR-01	Dept. of Quality Control Lab
AM-140021-026	AM-140021-026-VP-01	AM-140021-026-VR-01	Dept. of Quality Control Lab

Recovery study

According to the requirement of SOP:AC-053 <Sampling and Testing Procedure for cleaning verification>, for the rinsing or reflexing cleaning: The recovery study on recent 10 compounds of the worst solubility for Asymchem TJ4 has been finished, and all the results are more than 50%, so the 50% is defined as the criteria for Asymchem TJ4 (refer to TJ4-SR-2026-002< Study Report for Rinsing Recovery >). The criteria is applicable to ASYM-140021 project. All swab and rinse results will be corrected for recovery.

The analytical method validation experiments demonstrated that the selected analytical methods are capable of quantitatively detecting the target residues at levels meeting the established acceptance criteria. Target-specific methods were developed based on the established acceptance criteria and specific equipment information, and the LOQ concentrations were determined. The validation results confirmed that the LOQs meet the intended acceptance criteria.

Table 25, Recovery study results

Target residue	MOC	Recovery	RSD
ASYM-140852	Hastelloy	1% level: 71%	2%
		100% level: 87%	1%
		200% level: 87%	0%
Boc-L-Glu-OtBu (YL008618/ASYM-140413)	Hastelloy	5% level: 92%	3%
		100% level: 94%	4%
		200% level: 93%	3%

6.11.Dirty Hold Time Study

Dirty Hold time: The period and conditions of storage of equipment before cleaning. The purpose of Dirty Hold Time qualification is to demonstrate that residue ageing does not adversely affect the ability of the validated cleaning process to achieve defined visual and chemical acceptance criteria. The proposed DHT represents the maximum anticipated operational delay between batch completion and cleaning initiation. Validation data generated at this condition will therefore represent a worst-case challenge to the cleaning process.

Based on process historical knowledge and PAR study (Refer to TR-1400210305 <Technical Summary for LSN4182928 (ASYM-140021) Step 1 PAR Study> (current version), TR-1400210306< Technical Summary for LSN4182928 (ASYM-140021) Step 2 PAR Study >), the in-process intermediate ASYM-140851 is stable at 2~8ºC for 12 months and cleavage product ASYM-140852 is stable at -3~48ºC for 21d. Since the solid is stable at certain days and all of equipment/auxiliaries are at closed status at rest (solvent evaporation is minimal, which has minimal impact on equipment cleaning), the dirty hold time is currently set in **Table 26** as a guideline.

DIC/ DMF solution and Iodine/DMF solution are stable at room temperature under nitrogen blanket based on common chemical experience. Since the solution is stable at certain days and all of equipment/auxiliaries are at closed status at rest (solvent evaporation is minimal, which has minimal impact on equipment cleaning), the dirty hold time is currently set in **Table 26** as a guideline.

Table 26, Suggested current dirty hold time

Equipment ID	Suggested dirty hold time based on R&D study for manufacturing guide
TJ4S-1213-Sy16	7 days
TJ4S-1213-Sy16-R01	14 days
TJ4S-1200-AT01	14 days
TJ4S-1213-AT09	14 days

As more data from subsequent campaigns is collected, the dirty hold time shall be evaluated based on the production data, and an updated project specific dirty hold time shall be proposed. DHT qualification will be performed as part of the cleaning validation, where cleaning will be executed under the proposed maximum DHT condition and residue levels will be tested. If the validation results meet the acceptance criteria, this will demonstrate with actual data that ageing does not adversely affect cleaning effectiveness. This will be specifically discussed in the cleaning validation report.

For the equipment dirty hold time between batches during the continuous production, all the data for the dirty hold time will be summarized in the cleaning validation report after the cleaning validation, and a specific dirty hold time for this project will be proposed in the report.

During PPQ batch and subsequent commercial manufacturing, manufacturing data shall be collected to further justify the dirty hold time. Should this time limit be exceeded, an investigation shall be initiated and an assessment shall be conducted.

For dirty hold time data collected during manufacturing, two types of dirty hold time data are collected: One is the time between the equipment last use and the start time of cleaning, another is time between the equipment last use and next consecutive use without cleaning.

Two key hold times shall be picked from the subsequent production, that is: the 3rd longest dirty hold time and the 1st longest dirty hold time, further conclude below.

1. If the dirty hold time is equal or less than the 3rd longest dirty hold time, it is regarded that the equipment was safe and no need cleaning, the equipment could be used in the later continuous production.
2. If the "hold time" is demonstrated/verified for one or two times during cleaning validation or subsequent commercial production, the hold time is regarded as alert dirty hold time (still safe since this hold time was demonstrated in the previous batches), and shall pay close attention to the impurity profile of the finished product. Any atypical impurity or OOS to the material specification shall be investigated.
3. If the "hold time" is never demonstrated/verified during cleaning validation or subsequent commercial production, the hold time is regarded as risk dirty hold time. Then shall evaluate the impurity profile of the finished product. Any atypical impurity or OOS to the material specifications should be investigated. If this hold time is proved acceptable, then this dirty hold time should be updated per this new outcome.

6.12.Clean Hold Time

Clean Hold Time: The time between cleaning and equipment reuse, prior to additional cleaning.
The clean hold time is defined as NMT 6 months in the GMP facility after the equipment is cleaned and dried. Considering the environment control and equipment storage is same between the GMP facility in TJ4 site and the GMP facility in the site for CHT validation, hence the CHT with NMT 6 months is applicable for the GMP facility in TJ4 site after equipment cleaning (refers to VR-180.01-02.03 <Validation Report of Clean Hold Time (6 months)>). If outside of this time, the equipment shall be cleaned again before next use.

7 Product distribution plan during the cleaning validation

During the cleaning validation, the impacted batch involved in the cleaning validation shall be released according to the agreement between Asymchem and customer before the cleaning validation is executed completely. However, there will be interim cleaning validation report(s) to summarize the current cleaning validation results and indicate no impact on the product quality.
Justification for concurrent cleaning validation shall be approved by the site quality head.
QA shall release the TJ4S-1213-Sy16 equipment for the next batch only after the visual and analytical results of the run meet the acceptance criteria.
The interim cleaning validation report shall be approved prior to release of the Step 3 product.

8 Impurity profile summary and impurity trending analysis

To further demonstrate the cleaning validation strategy feasible, the impurity profile summary from COA release test results will be collected and impurity trending analysis will be conducted by using control chart. Also, the corresponding impurity evaluation information will be summarized in the cleaning validation report. Cleaning validation forms one component of the overall contamination-control strategy. Mechanisms for continuously assuring cleaning effectiveness, including review of product quality data, impurity trend analysis, deviation investigations, change control and periodic assessment of cleaning process performance.

9 Amendment of protocol

Any amendment of protocol during execution of protocol will shall be evaluated and approved.

10 Deviation and Change Handling

1) Any deviations found during execution of the cleaning validation shall be recorded and investigated according to QA-010 <Deviation Investigation> (current version). Any result exceeding an established acceptance criterion shall be considered a cleaning validation failure unless investigation demonstrates an assignable cause unrelated to cleaning effectiveness. Failure investigations shall be conducted in accordance with QA-010 <Deviation Investigation> (current version), covering root cause analysis, impact assessment and corrective actions. The investigation shall clearly distinguish between failure causes related to cleaning effectiveness and those unrelated to cleaning effectiveness (e.g., analytical method error, sampling error, instrument malfunction, etc.). If an assignable cause unrelated to cleaning effectiveness cannot be demonstrated, the failure shall be considered a cleaning process failure, and an assessment shall be performed to determine whether re-cleaning, re-validation or other corrective actions are required. Failures and their investigation conclusions shall be documented in a deviation report and assessed prior to protocol closure.
2) Any changes executed during the cleaning validation shall be recorded and approved according to QA-006<change control> (current version).
3) All copies of deviation investigation and changes control shall be attached to cleaning validation report. The information summary of deviation and changes control shall be summarized in **Annex 3** and **Annex 4**. Any deviation or change shall be formally assessed for its impact on protocol validity, data integrity and cleaning process qualification prior to protocol closure. The assessment shall involve the Quality department and be documented in a deviation report or change control record. The assessment shall clearly determine whether the deviation or change affects the cleaning process qualification status and whether corrective actions or additional verification are required. Prior to protocol closure, it shall be confirmed that all deviations and changes have been closed or assessed.

11 Action taken when out-of-limit result found

1) When the result of residue testing is out of limit, the results shall be recorded and investigation shall be performed according to QA-010 <Deviation Investigation> (current version).

2) If any significant unknown impurity peaks are found after comparing the blank of rinsing/refluxing/swabbing with sample solution, each unknown impurity shall be calculated as the target residual compound, respectively, and list the test results of all the unknown impurities and fill the maximum result for each unknown impurity in the corresponding cleaning verification test request.

When performing analytical quantitative testing using chromatography method, if the maximum result of a single unknown impurity is greater than the acceptable residue limit, laboratory investigation shall be initiated according to QA-010 <Deviation Investigation> (current version). If no assignable cause is found in the laboratory investigation, production investigation shall be initiated.

12 Summary and Conclusion

When the protocol and related tests are finished, the relative validation report shall be completeted in time. The final report shall include the following contents:

1) Approval page
2) Evaluation of test results
3) Conclusion / comments
4) Deviation/Changes control list and copies.

Successful completion of this protocol will provide documented evidence that the approved cleaning process consistently removes process residues from equipment used in the manufacture of ASYM-140851 Step 1 and maintains equipment in a condition suitable for subsequent manufacture. Successful completion of three consecutive cleaning validation runs meeting all predefined acceptance criteria will establish the qualified state of the cleaning process and support implementation into routine commercial manufacture.

13 Revalidation

Revalidation shall be considered under the following circumstances:

- Equipment change or major modification to existing equipment
- Cleaning procedure change
- Cleaning solvent change
- Evidence of cleaning failures (batch failures, cross contamination, routine monitoring)
- Major change on the process

14 References

(1) Cleaning Validation Master Plan (CVMP-01)
(2) Cleaning Process Design and Development Report
(3) Approved cleaning methods (CMa-TJ414002101, CMb-TJ414002101)
(4) OEB Assessment Report
(5) Analytical Method Validation Report

(6) Validation Report of Clean Hold Time (VR-180.01-02.03)

(7) Relevant SOPs (DS-GMP-003<Equipment cleaning management procedure for GMP production (DS)>, DS-GMP-043<Cleaning Validation (DS)>, QA-010<Deviation Investigation>, QA-006<Change Control>, QAP-005<Quality Risk Management Policy>)

(8) Process Validation Master Plan for Manufacturing of LSN4182928/ASYM-140021 (VP-14002101)

(9) EU GMP Volume 4 Annex 15: Qualification and Validation

(10) ICH Q7 Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients

(11) ICH Q9(R1) Quality Risk Management

(12) FDA Guidance for Industry: Process Validation, General Principles and Practices

(13) EMA Guideline on Setting Health-Based Exposure Limits for Use in Risk Identification in the Manufacture of Different Medicinal Products in Shared Facilities

Annex 1: Cleaning method list

Step	Cleaning Method No.	Attachment No.
1	CMa-TJ414002101 (current version)	Attachment 1 for CVP-1400210301.01
	CMb-TJ414002101 (current version)	Attachment 2 for CVP-1400210301.01

Annex 2: PI&D diagram for the swab sampling locations

Equipment	PI&D number	Attachment No.
Peptide solid phase synthesizer	Sampling location Diagram for Peptide Solid Synthesizer	Attachment 3 for CVP-1400210301.01

Annex 3: Deviation record

Step/Affected Batches	Deviation Description	Root Cause	Date	Quality impact	Deviation Document No.
-	-	-	-	-	-
-	-	-	-	-	-

Annex 4: Change record

Step/Affected Batches	Change description	Date	Quality impact	Change Document No.
-	-	-	-	-
-	-	-	-	-

Annex 5: The definition of abbreviation

Abbreviation	Description
ARL	Allowable Residue Limit
MACO	Maximum Acceptable Carry Over
MOC	Material of Construction
MBS	When Minimum batch size of next product(s) is known, MBS is selected as Minimum batch size of next product(s). When Minimum batch size of next product(s) is unknown, MBS is selected as Minimum batch size of cleaned equipment.
MTDDnext	Standard maximum therapeutic daily dose of the next product If the next product is not daily dose, it needs transfer into daily dose.
MTDD previous	Standard minimum therapeutic daily dose of the previous product

Abbreviation	Description
	If the previous product is not daily dose, it needs transfer into daily dose.
SF	Safety factor According to observation and statistics of clinical pharmacology, toxicology and application, there will be no pharmacological reaction of 1/1000 MTDD for very sensitive bodies. For safety, take 1000 as safety factor for calculation is acceptable.
PDE	Permitted Daily Exposure
OOS	Out of specification
ATR	Atypical result
NMT	No more than

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